

Network Layer in IoT Communications

Internet of Things



CS5055 - 2025I
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SRC: KUROSE – COMPUTER NETWORKS BOOK
PETERSON AND DAVIE – COMPUTER NETWORKS BOOK

Executive Summary

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- **Motivation:** What is the role of the network layer for IoT applications?
- Topics:
 - Network layer Fundamentals.
 - IoT challenges.
 - IP vs Non-IP protocols
 - LoRaWan

Outline

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Fundamentals

IoT challenges

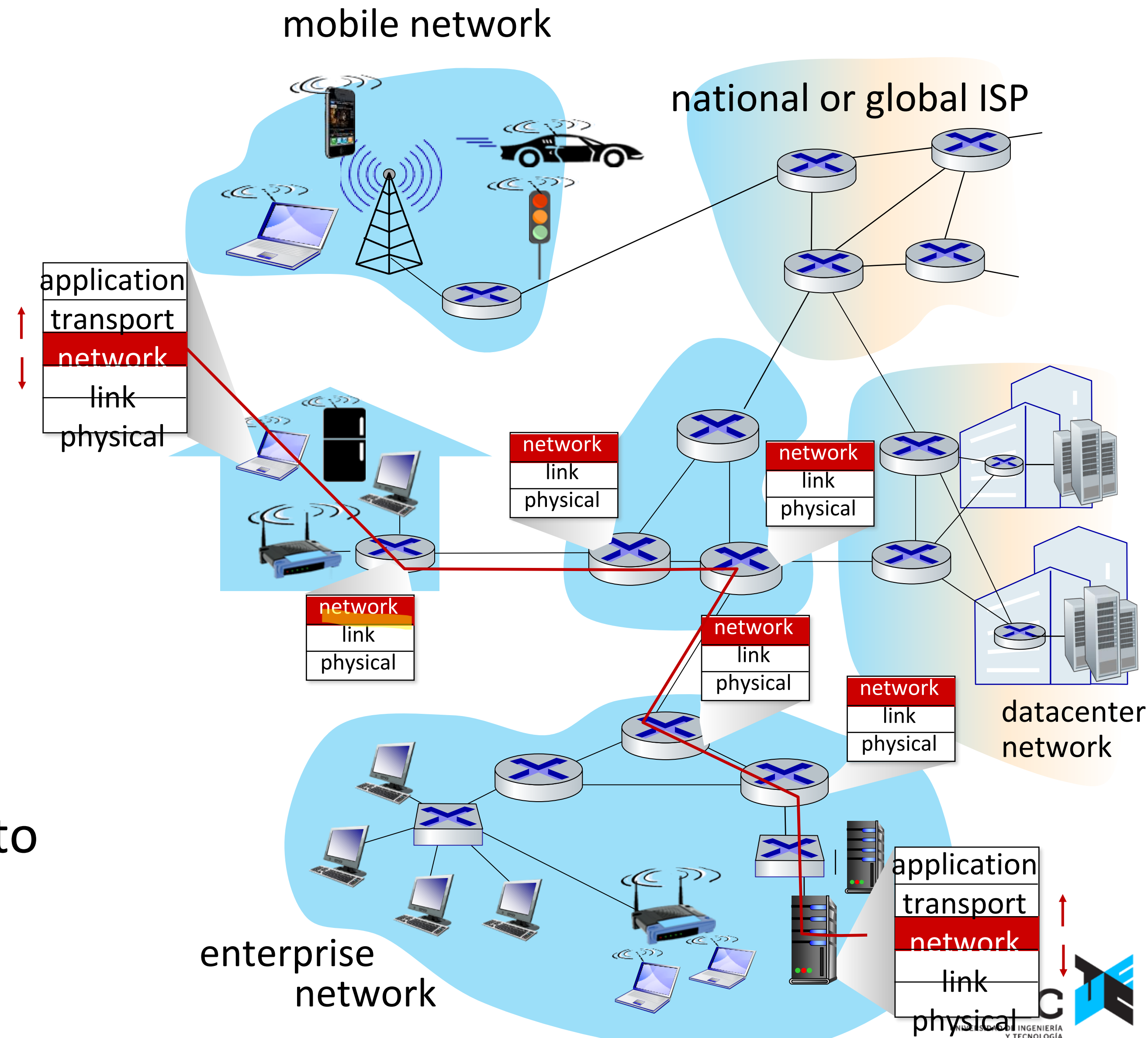
IP vs. No-IP en IoT

LoRaWan

Conclusion

Network-layer services and protocols

- transport segment from sending to receiving host
 - **sender:** encapsulates segments into datagrams, passes to link layer
 - **receiver:** delivers segments to transport layer protocol
- network layer protocols in *every Internet device*: hosts, routers
- **routers:**
 - examines header fields in all IP datagrams passing through it
 - moves datagrams from input ports to output ports to transfer datagrams along end-end path



Two key network-layer functions

network-layer functions:

- *forwarding*: move packets from a router's input link to appropriate router output link
- *routing*: determine route taken by packets from source to destination
 - *routing algorithms*

analogy: taking a trip

- *forwarding*: process of getting through single interchange
- *routing*: process of planning trip from source to destination

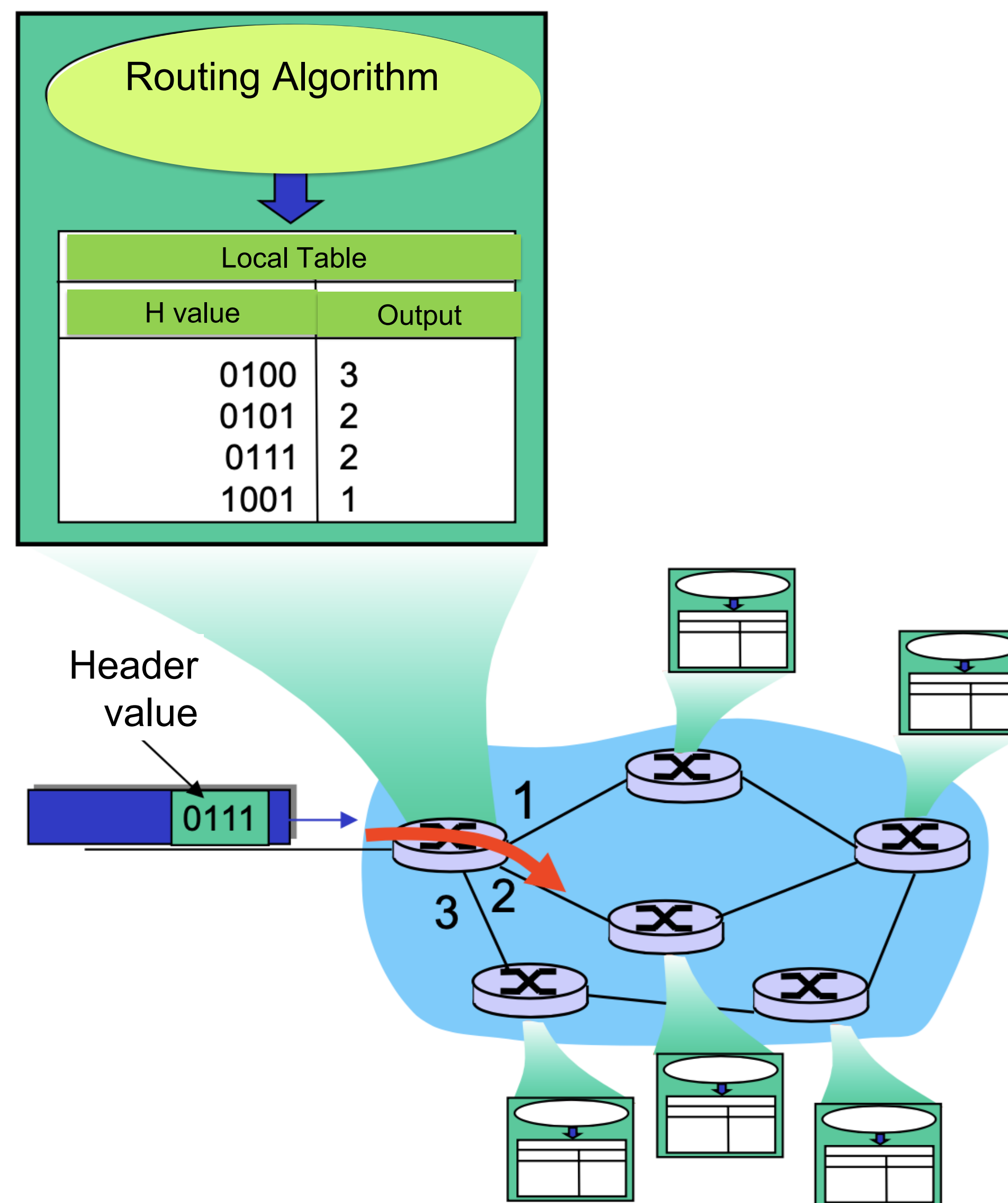


forwarding



routing

Two key network-layer functions

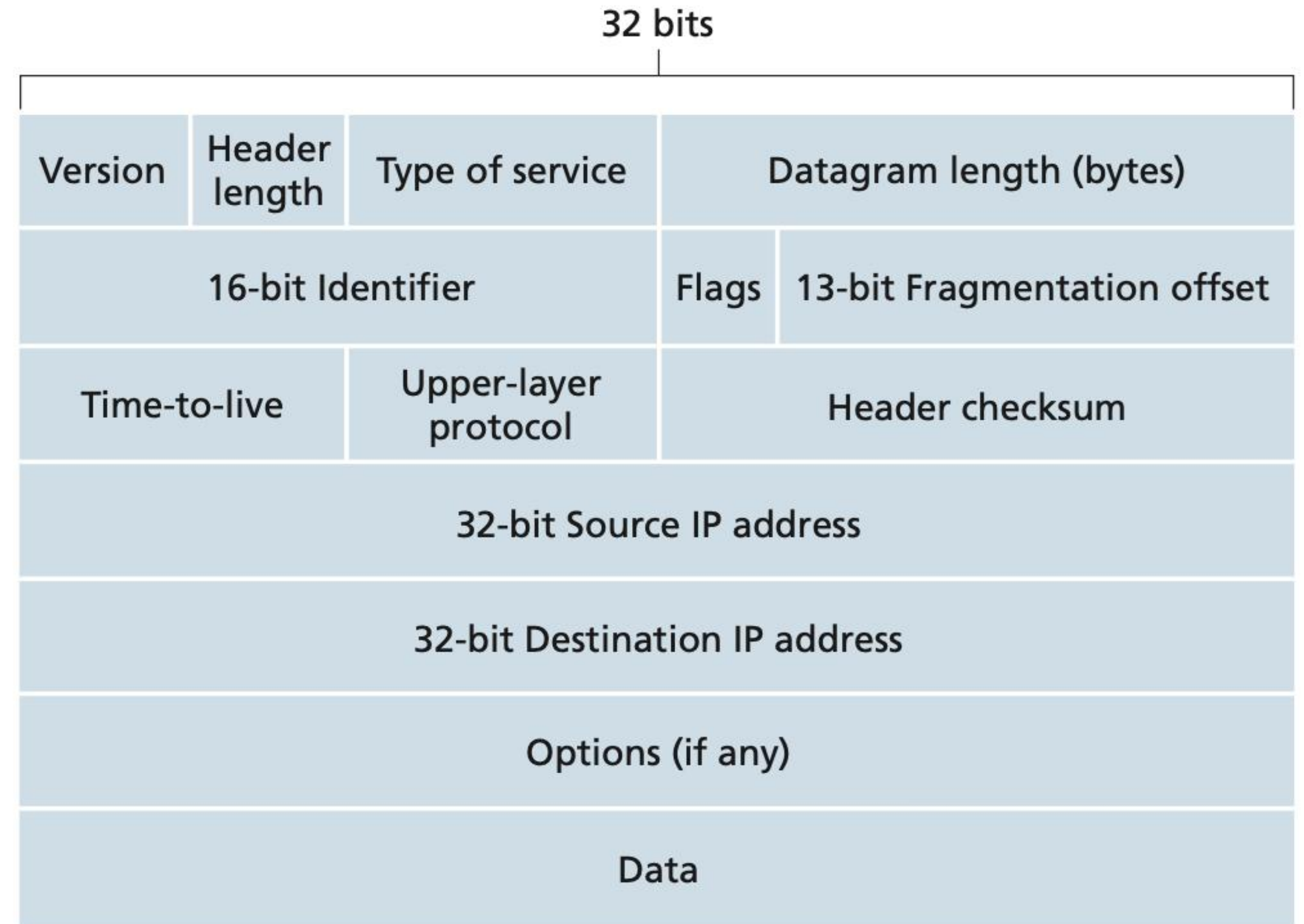


IP Datagram

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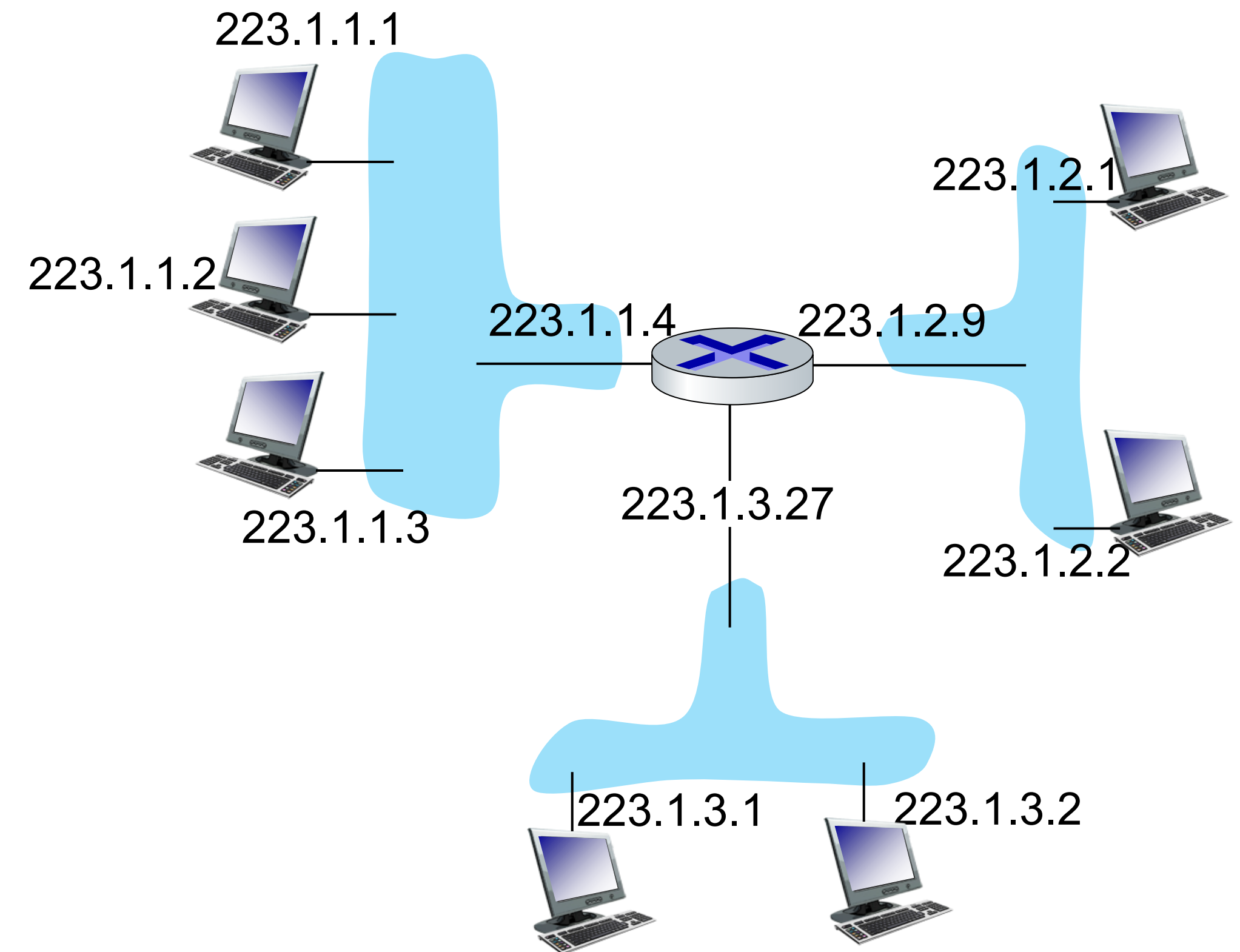
IP Overhead

- 20 bytes from TCP
- 20 bytes from IP
- = 40 bytes + overhead from application layer



IP addressing: Introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **interface:** connection between host/router and physical link
- router's typically have multiple interfaces
- host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

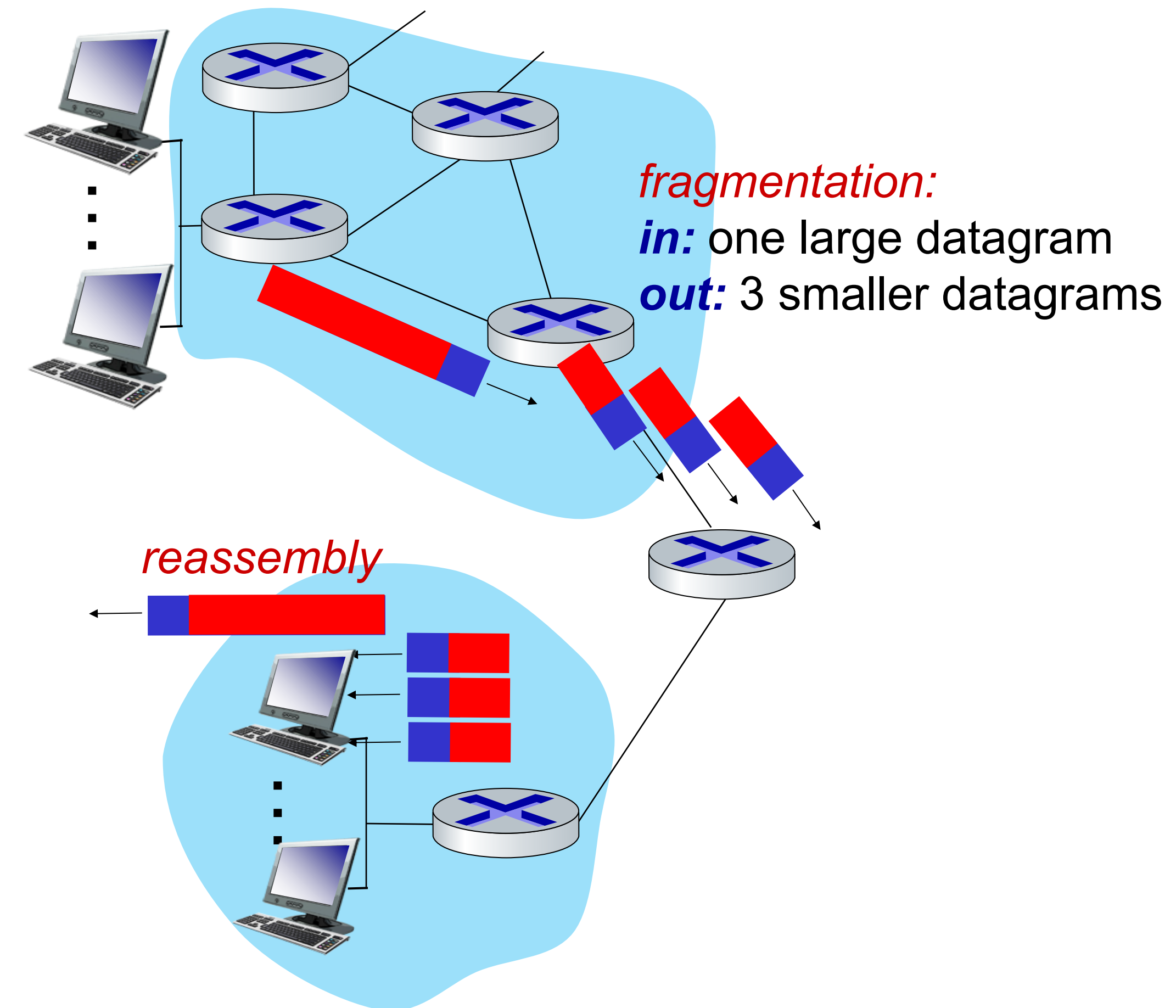
223.1.1.1 = 11011111 00000001 00000001 00000001

223
1
1
1

IP fragmentation/reassembly

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- network links have MTU (max. transfer size) - largest possible link-level frame
 - different link types, different MTUs
- large IP datagram divided (“fragmented”) within net
 - one datagram becomes several datagrams
 - “reassembled” only at *destination*
 - IP header bits used to identify, order related fragments



IP fragmentation/reassembly

example:

- 4000 byte datagram
- MTU = 1500 bytes

	length	ID	fragflag	offset		
	=4000	=x	=0	=0		

*one large datagram becomes
several smaller datagrams*

1480 bytes in
data field

offset =
 $1480/8$

	length	ID	fragflag	offset		
	=1500	=x	=1	=0		

	length	ID	fragflag	offset		
	=1500	=x	=1	=185		

	length	ID	fragflag	offset		
	=1040	=x	=0	=370		

Outline

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Fundamentals

IoT challenges

IP vs. No-IP en IoT

LoRaWan

Conclusion

Scalability in IoT

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- **a) Device Addressing**

IPv4 is insufficient: Only allows ~4.3 billion addresses (exhausted since 2019).

Ex: A network of 100,000 sensors in a city would require complex NAT and artificial hierarchies.

- **b) Network Overload**

Broadcast storms: Traditional protocols (e.g., ARP) generate excessive traffic when scaling.

Ex: 10,000 devices sending discovery messages would crash the network.

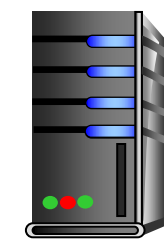
- **c) Backend Processing**

Centralized servers can become bottlenecks with millions of devices sending data simultaneously.

DHCP client-server scenario

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DHCP server: 223.1.2.5



DHCP discover

Broadcast: is there a
DHCP server out there?

Arriving client



DHCP offer

Broadcast: I'm a DHCP
server! Here's an IP
address you can use

DHCP request

Broadcast: OK. I would like
to use this IP address!

DHCP ACK

Broadcast: OK. You've got
that IP address!

The two steps above can be
skipped "if a client
remembers and wishes to
reuse a previously allocated
network address" [RFC 2131]

Scalability in IoT

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- **IPv6 and 6LoWPAN**

IPv6 enables addressing for **~340 undecillion devices**, eliminating the need for NAT.

6LoWPAN compresses IP headers to reduce bandwidth usage in constrained networks.

- **Network Hierarchies**

Gateways (e.g., in LoRaWAN) aggregate and pre-process data before sending it to the cloud, reducing traffic load.



Practical Example:

Scenario: A city deploys **10,000 smart streetlights**.

Without IPv6: Requires complex NAT and subnetting.

With 6LoWPAN: Each streetlight gets a **unique IPv6 address** and communicates directly.

Energy Efficiency in IoT

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- **a) High Power Consumption in Traditional Protocols**

TCP/IP overhead: Handshakes, acknowledgments, and retransmissions drain battery life.

Ex: A Wi-Fi sensor sending small data packets wastes energy on protocol overhead.

- **b) Duty Cycling Limitations**

Devices must **sleep** most of the time but wake up periodically to transmit.

Trade-off: Longer sleep = lower power, but may miss critical events.

- **c) Energy-Hungry Wireless Technologies**

Cellular (4G/5G/NB-IoT): High power for long-range communication.

Wi-Fi/Bluetooth: Short-range but inefficient for sporadic transmissions.

Energy Efficiency in IoT

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- **Low-Power Wide-Area Networks (LPWAN)**

LoRa/LoRaWAN:

Ultra-low power (10+ years on a single battery).

Adaptive Data Rate (ADR) optimizes transmission power.

Sigfox/NB-IoT:

Designed for small, infrequent transmissions.

- **Optimized Communication Protocols**

MQTT-SN (MQTT for Sensor Networks):

Lightweight version of MQTT, removes TCP overhead.

CoAP over UDP:

No connection setup, reduces energy by 90% vs HTTP.

Scheduled Transmissions (LoRaWAN Class B):

Devices wake up **only at fixed intervals** to save power.



Practical Example:

Scenario: A wildlife conservation project deploys 500 battery-powered sensors across a rainforest to monitor: Temperature, humidity, Animal movement.

Wi-Fi-based sensors: Require constant power, batteries drain in weeks, high maintenance costs in remote areas.

LoRaWAN + Ultra-Low-Power Design:

Sensors transmit 1x per hour (10ms LoRa bursts @ 30mA), Sleep mode between transmissions (0.1 μ A power draw).

Next meeting: Outline

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Introduction

Addressing IPv4

IP vs. No-IP en IoT

LoRaWan

Conclusion

IP Based Protocols

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- End-to-End Connectivity: Devices communicate directly with cloud/enterprise systems.
- Interoperability: Works with existing IT infrastructure (routers, firewalls).
- Standard Security: TLS/DTLS, OAuth 2.0, and IPSec support.
- Scalability: IPv6 ensures unlimited address space

IP Based Protocols

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IPv6 & 6LoWPAN

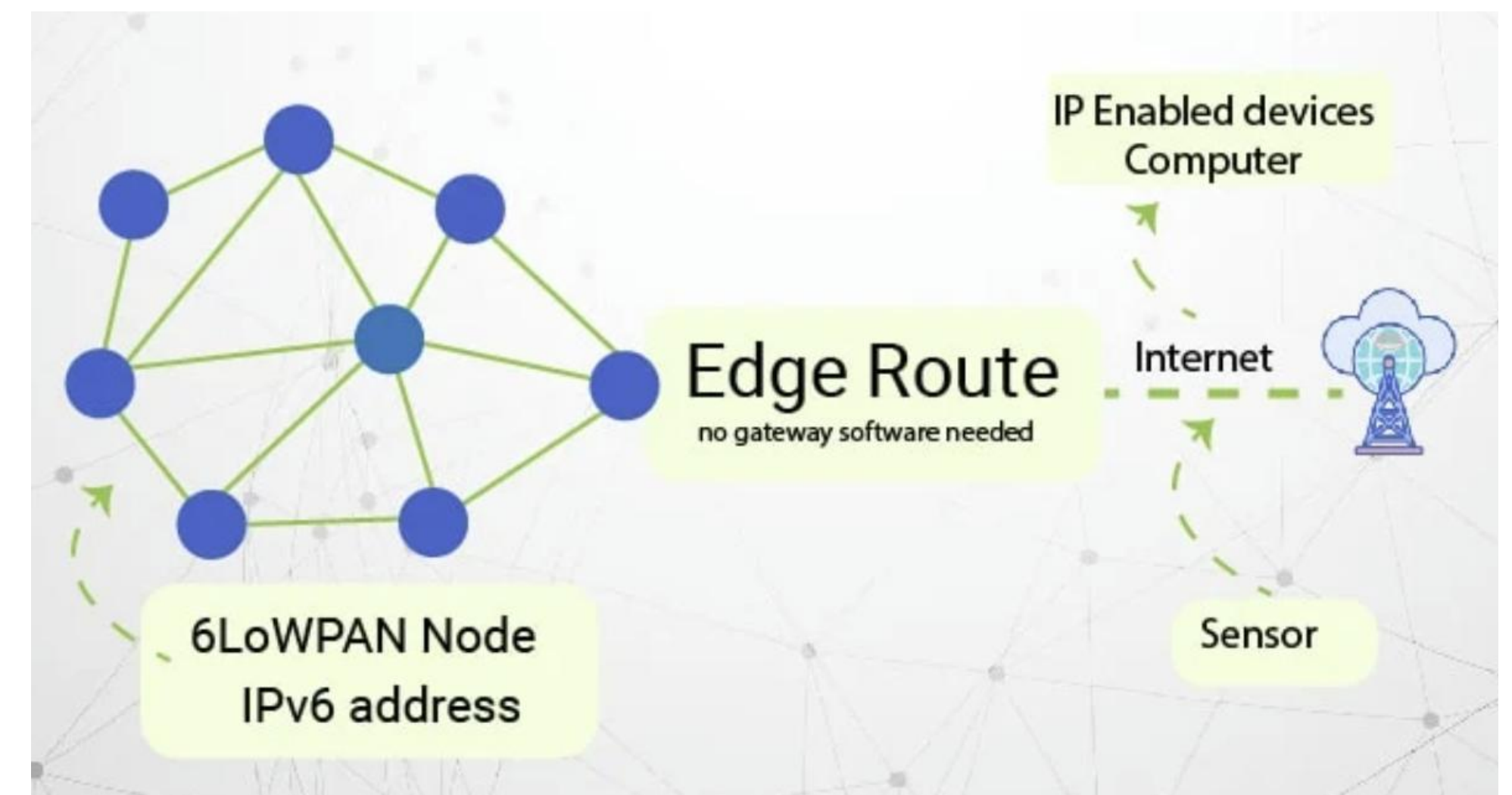
Purpose: Enable IP connectivity for low-power, constrained devices.

- IPv6: Provides 340 undecillion unique addresses, eliminating NAT.
- 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks):
 - Compresses IPv6 headers (from 40 bytes to 2–4 bytes).
 - Supports fragmentation for networks with small MTU (e.g., IEEE 802.15.4).

Use Cases:

Smart meters (e.g., AMI systems).

Industrial sensors (wired/wireless hybrid networks).



IP Based Protocols

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CoAP (Constrained Application Protocol)

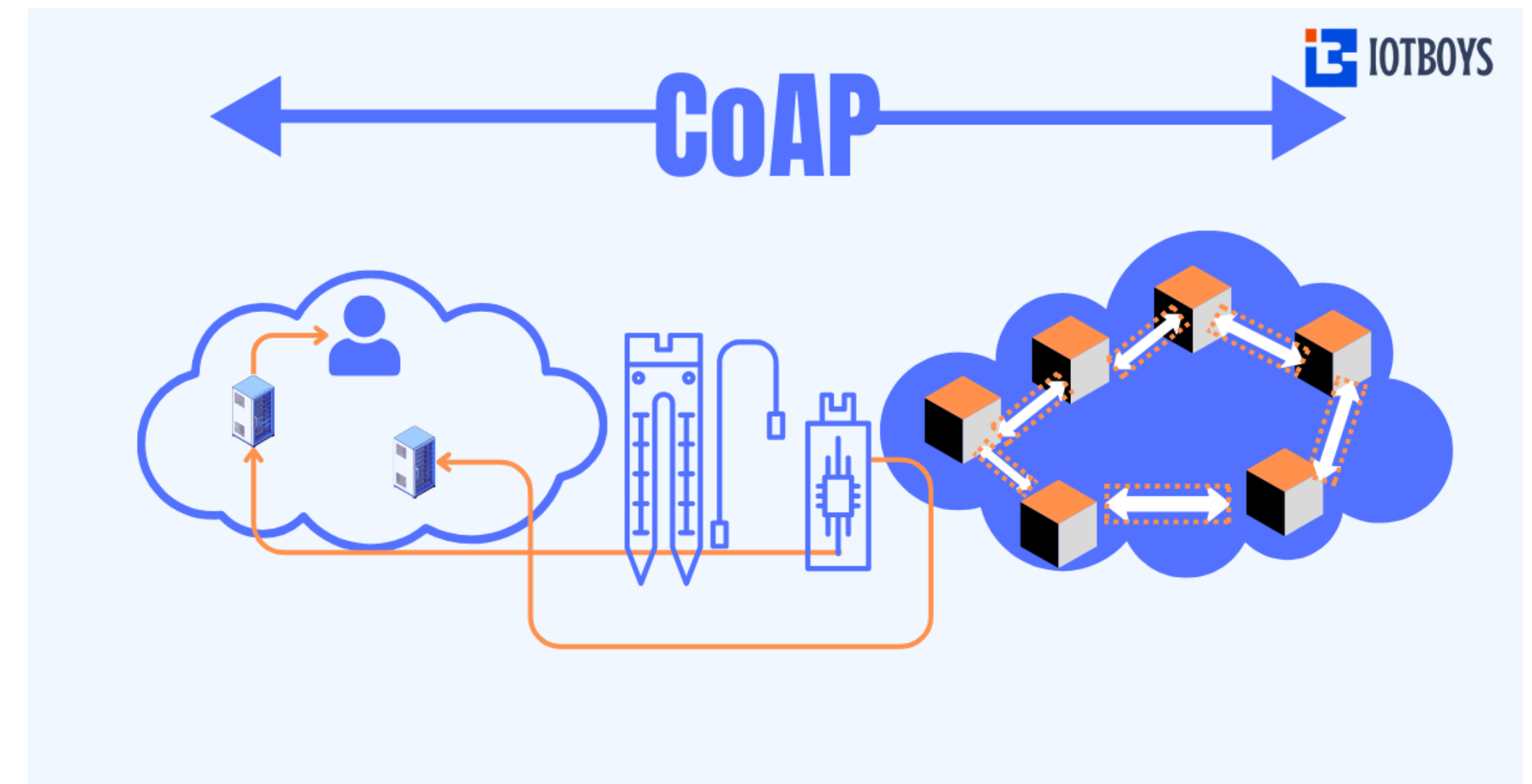
Purpose: Lightweight alternative to HTTP for resource-constrained devices.

- Runs over **UDP** (no TCP handshake overhead).
- Supports **multicast**, observation (server push), and DTLS encryption.

Use Cases:

Smart lighting (e.g., **Philips Hue** uses CoAP over Zigbee IP).

Wearables (e.g., firmware updates via CoAP).



Non-IP Based Protocols

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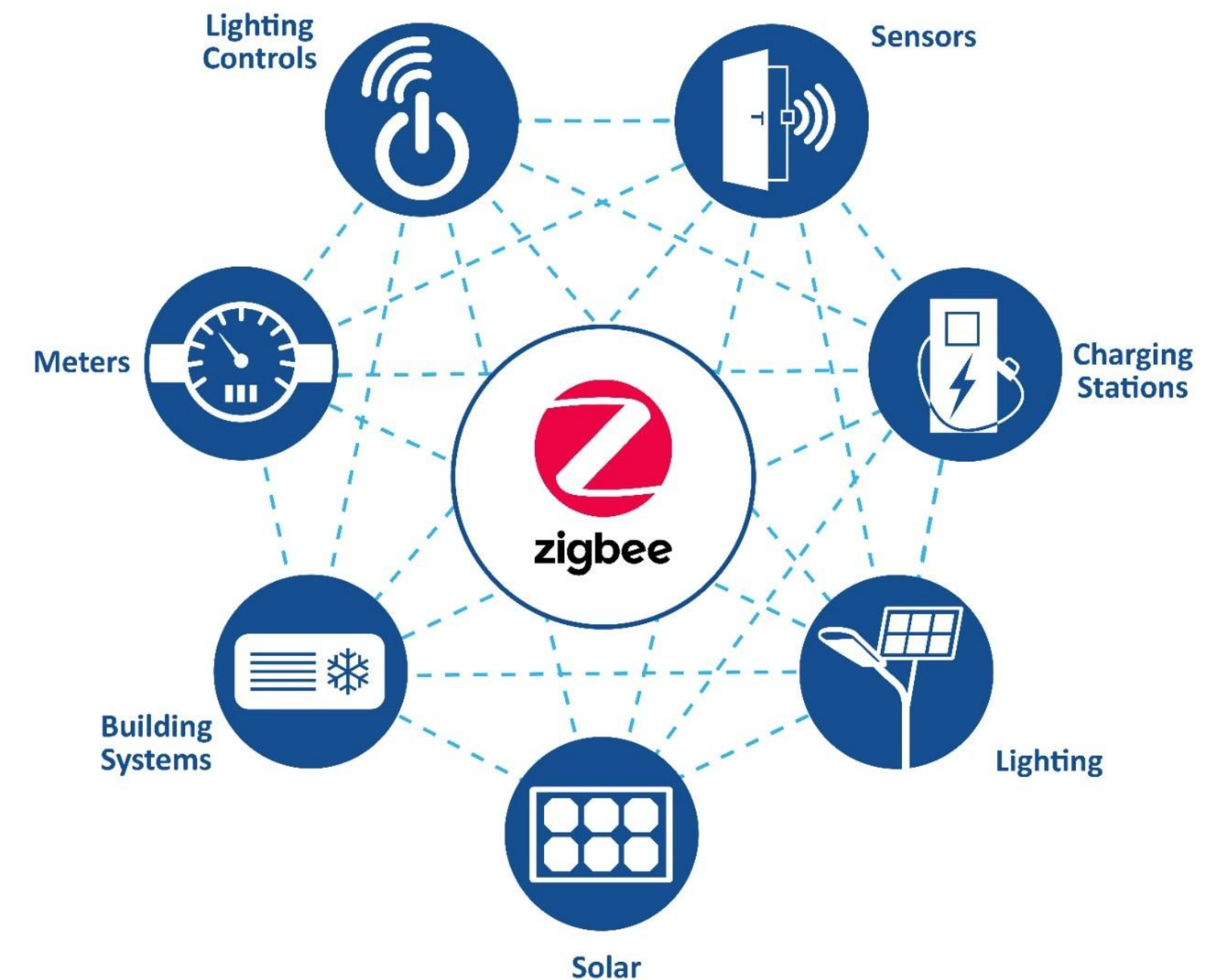
Zigbee

Purpose: Short-range mesh networking for smart homes/industries.

- **Non-IP Mesh:** Uses **16-bit short addresses** (managed by a coordinator).
- **Protocols:**
 - Zigbee PRO** (large-scale networks).
 - Zigbee 3.0** (unifies all Zigbee profiles).

Use Cases:

Home automation (Philips Hue, Amazon Echo).
Industrial sensor networks.



Non-IP Based Protocols

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LoRaWAN

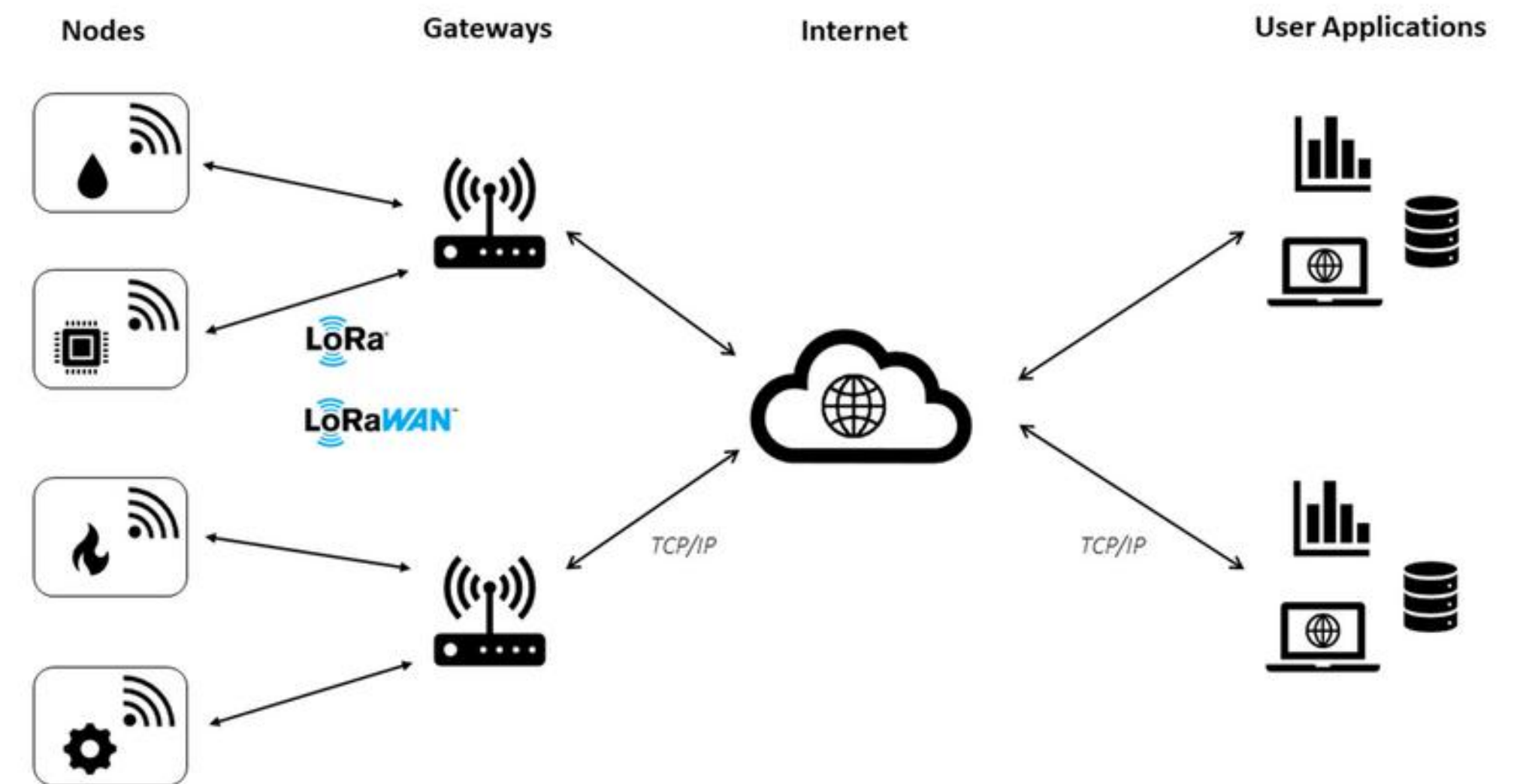
Purpose: Long-range, low-power communication for battery-operated devices.

- **No IP Addressing:** Uses **DevEUI/AppEUI** (64-bit unique identifiers).
- **Network Architecture:**
End Nodes → Gateways → Network Server → Application Server.

Use Cases:

Smart agriculture (soil sensors).

Asset tracking (no cellular coverage needed).



Outline

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Introduction

Addressing IPv4

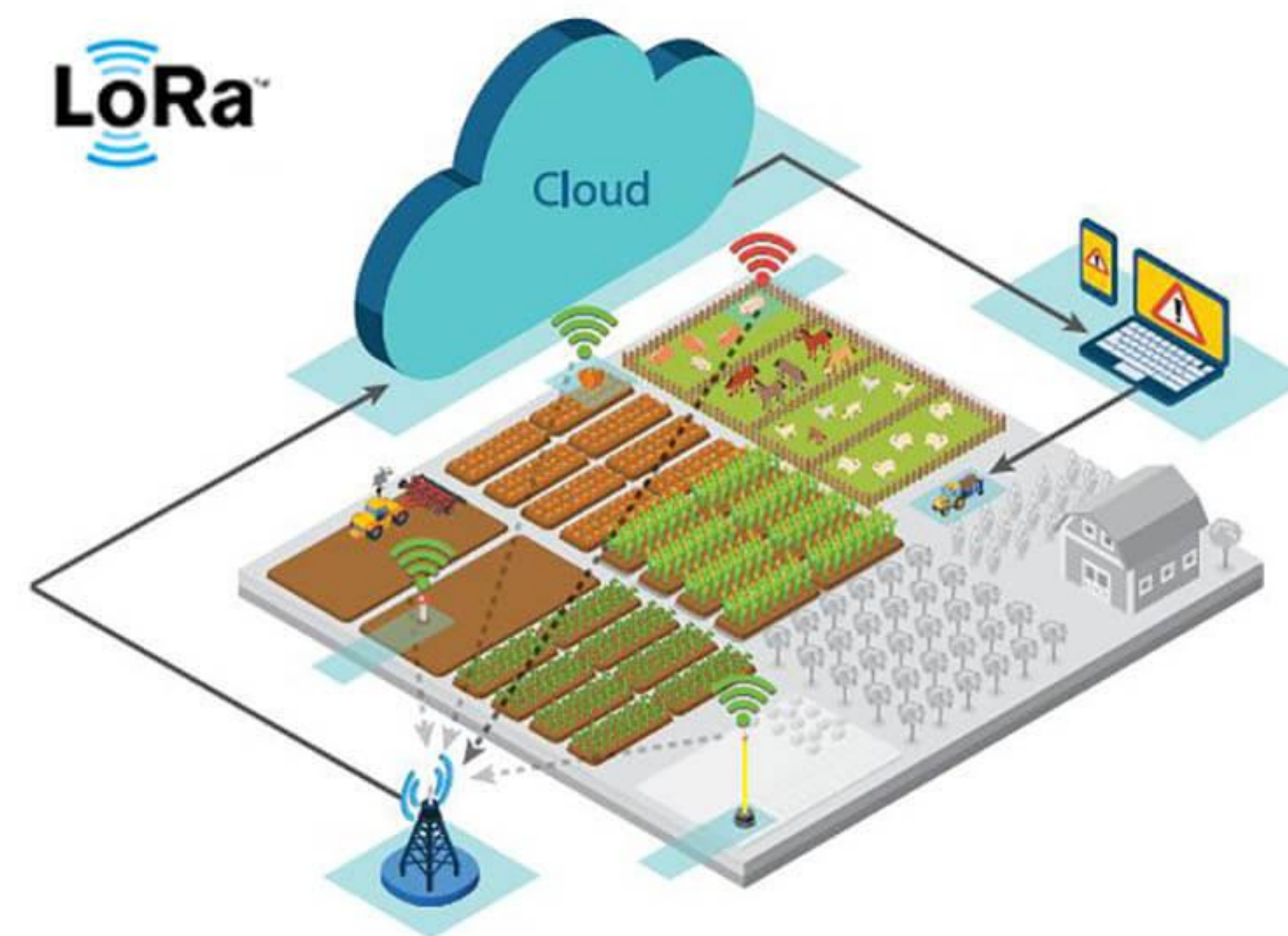
IP vs. No-IP en IoT

LoRaWan

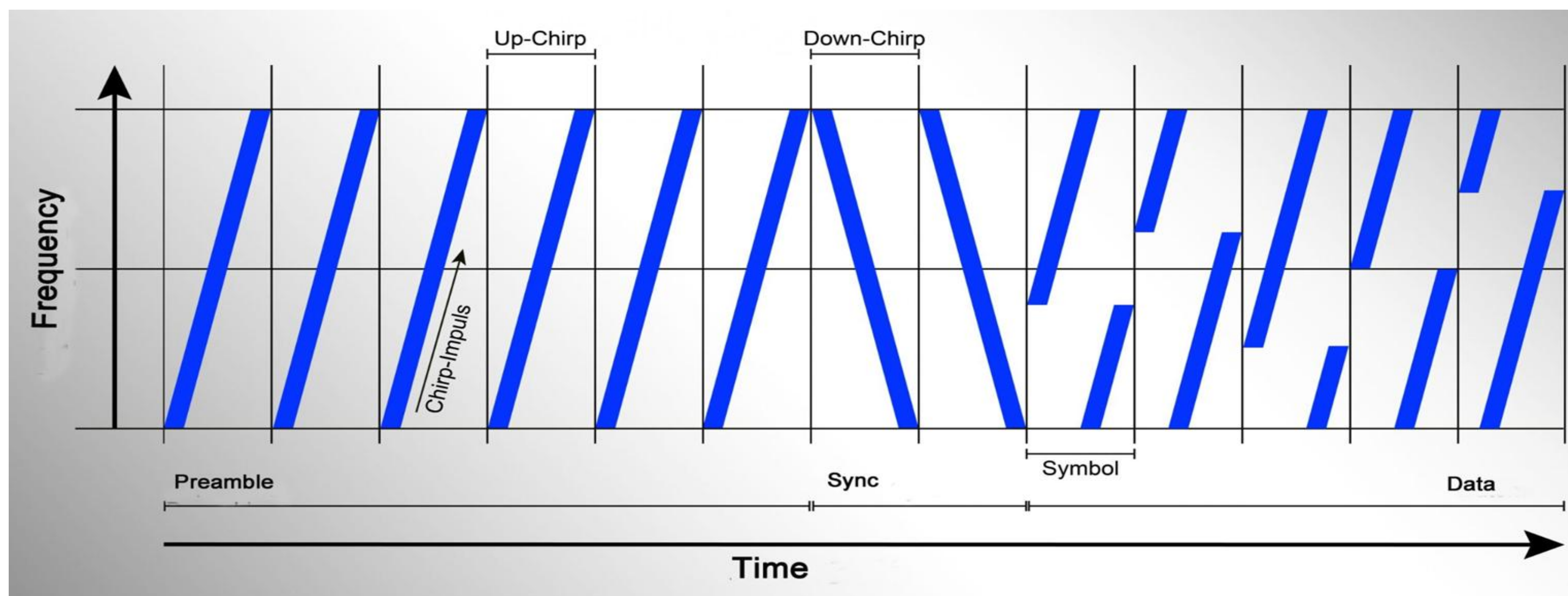
Conclusion

LoRa

Es una tecnología de modulación patentada por Semtech que permite la transmisión de datos de largo alcance (varios kilómetros) utilizando frecuencias de radio no licenciadas. LoRa se centra en la capa física de la comunicación inalámbrica y proporciona una forma eficiente de transmitir datos en entornos con interferencias y con un consumo de energía relativamente bajo.

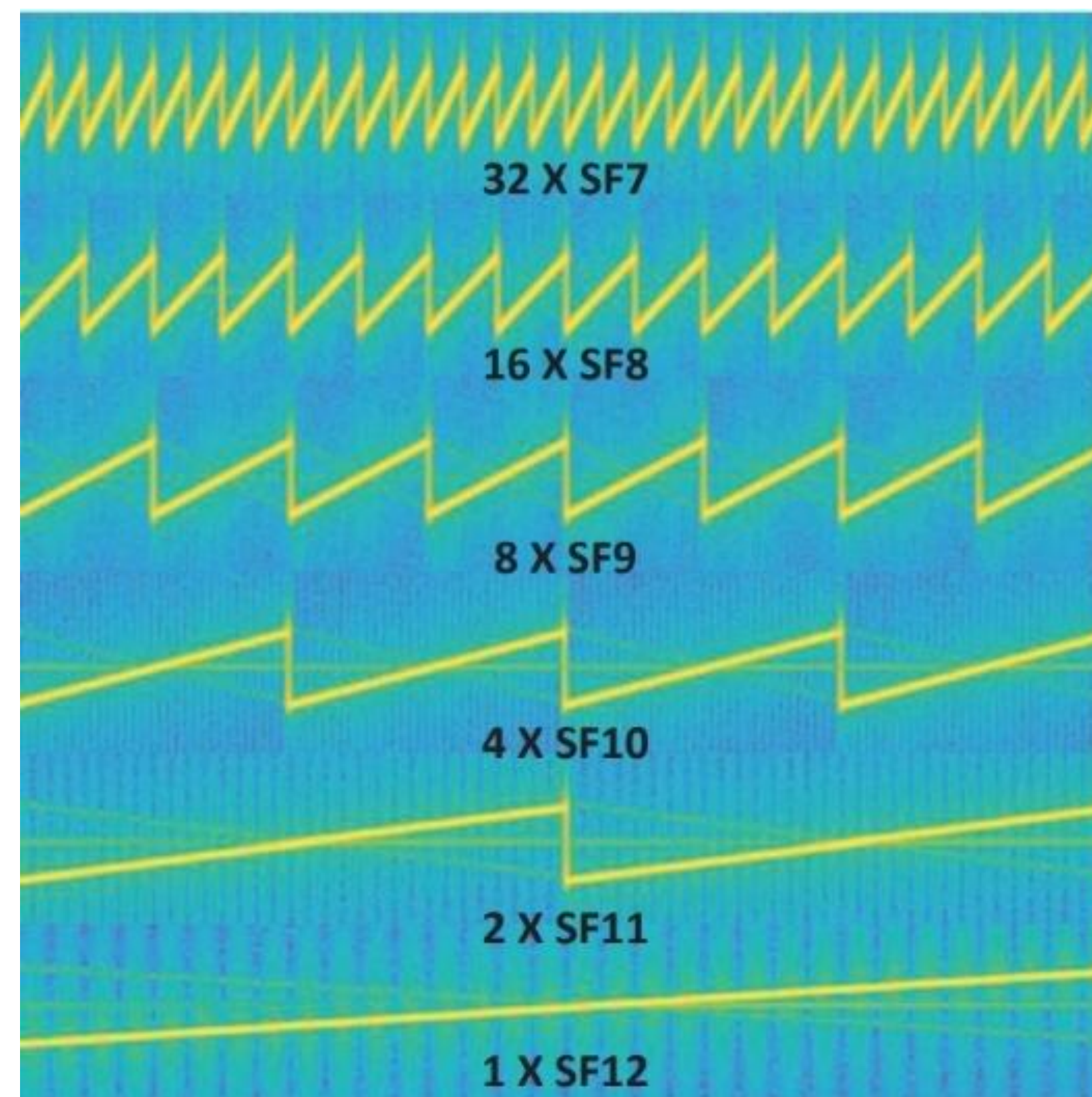


Chirp Spread Spectrum (CSS) : CSS es un método de modulación en el que los datos se codifican usando chirps, que son señales cuya frecuencia aumenta o disminuye linealmente con el tiempo. Cada chirp representa una unidad de información.



Parámetros asociados al CSS en LoRa:

- Spreading Factor (SF): Determina cuántos chirps se usan por símbolo. A mayor SF, mayor alcance y mayor robustez, pero menor velocidad de datos.
- Bandwidth (BW): Ancho de banda usado para la modulación.
- Coding Rate (CR): Cuánto de los datos transmitidos son redundancia para corrección de errores.



LoRa

Calcular el **tiempo de transmisión (ToA)** de un paquete LoRa usando la fórmula simplificada de $N_{payload}$ (con **CRC=0**, **H=0**, **DE=0**):

Payload (PL): 12 bytes.

Spreading Factor (SF): 8.

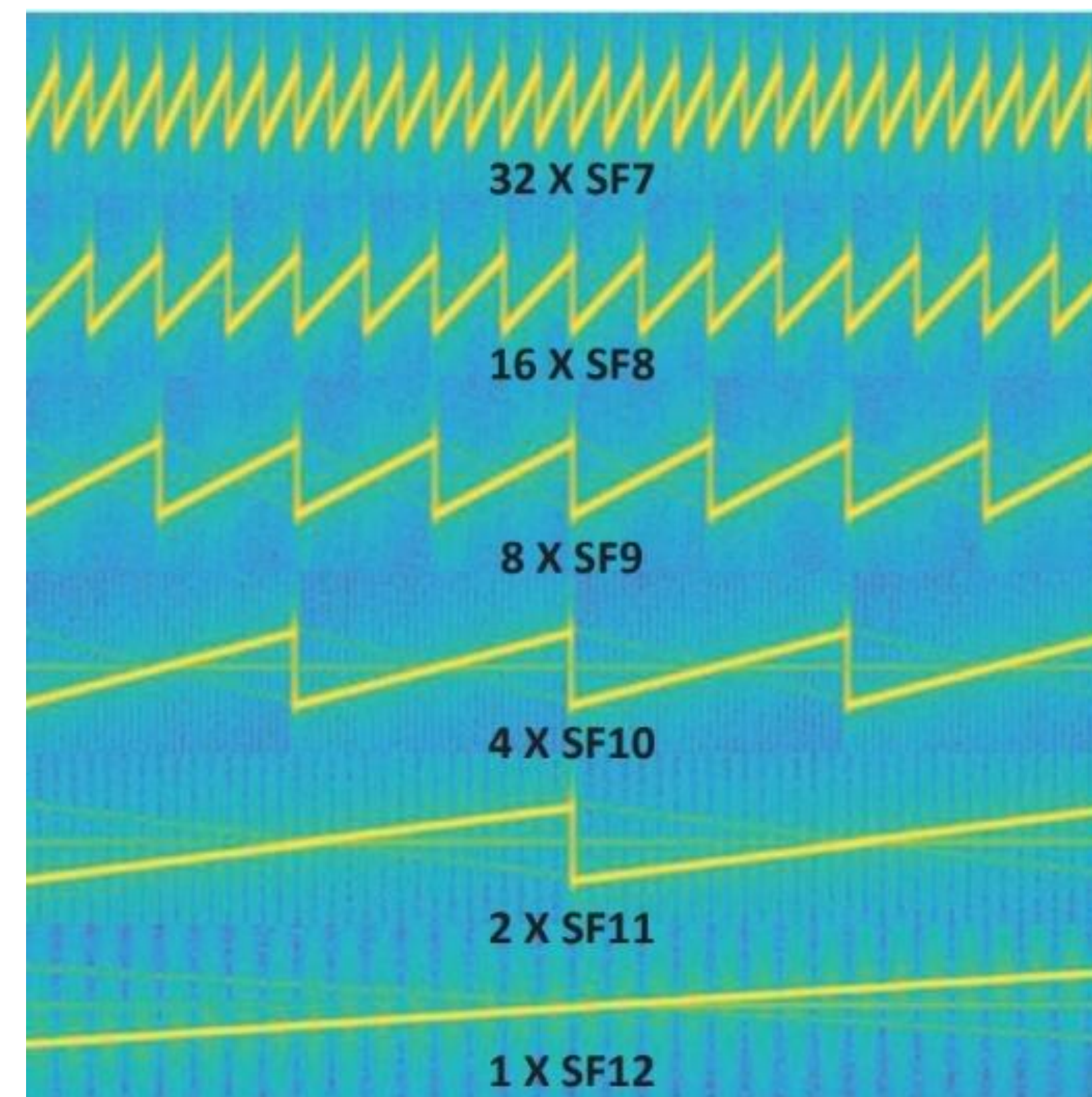
Bandwidth (BW): 125 kHz.

Coding Rate (CR): 1 (4/5).

Preamble Length: 8 símbolos (valor por defecto).

$$ToA = T_{preamble} + T_{payload}$$

$$T_{symbol} = \frac{2^{SF}}{BW}$$



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Calcular el **tiempo de transmisión (ToA)** de un paquete LoRa usando la fórmula simplificada de $N_{payload}$ (con **CRC=0**, **H=0**, **DE=0**):

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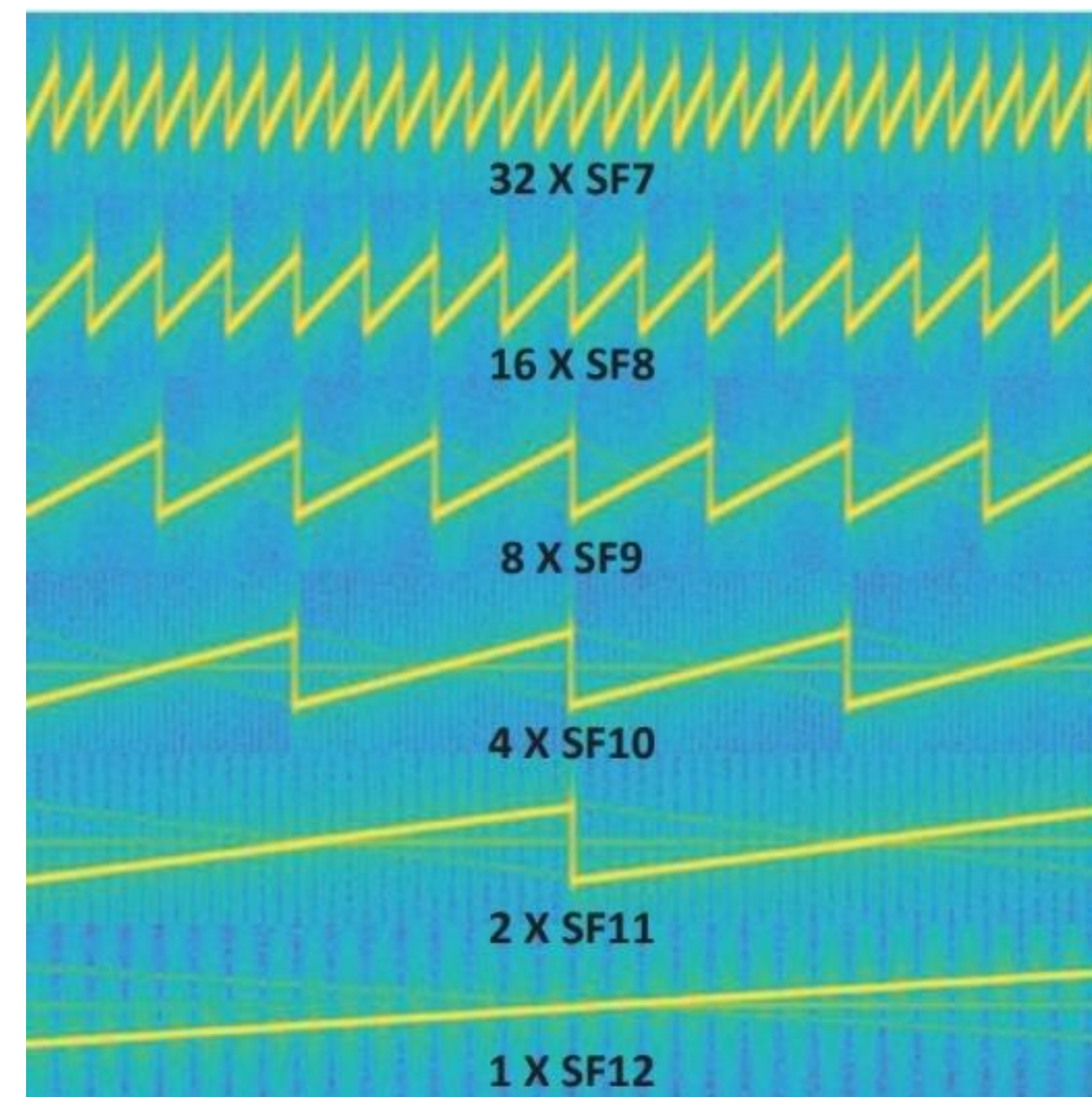
Spreading Factor (SF): 8.

Bandwidth (BW): 125 kHz.

Coding Rate (CR): 1 (4/5).

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$$T_{preamble} = (N_{preamble} + 4.25) \times T_{symbol}$$



LoRa

Calcular el **tiempo de transmisión (ToA)** de un paquete LoRa usando la fórmula simplificada de $N_{payload}$ (con **CRC=0**, **H=0**, **DE=0**):

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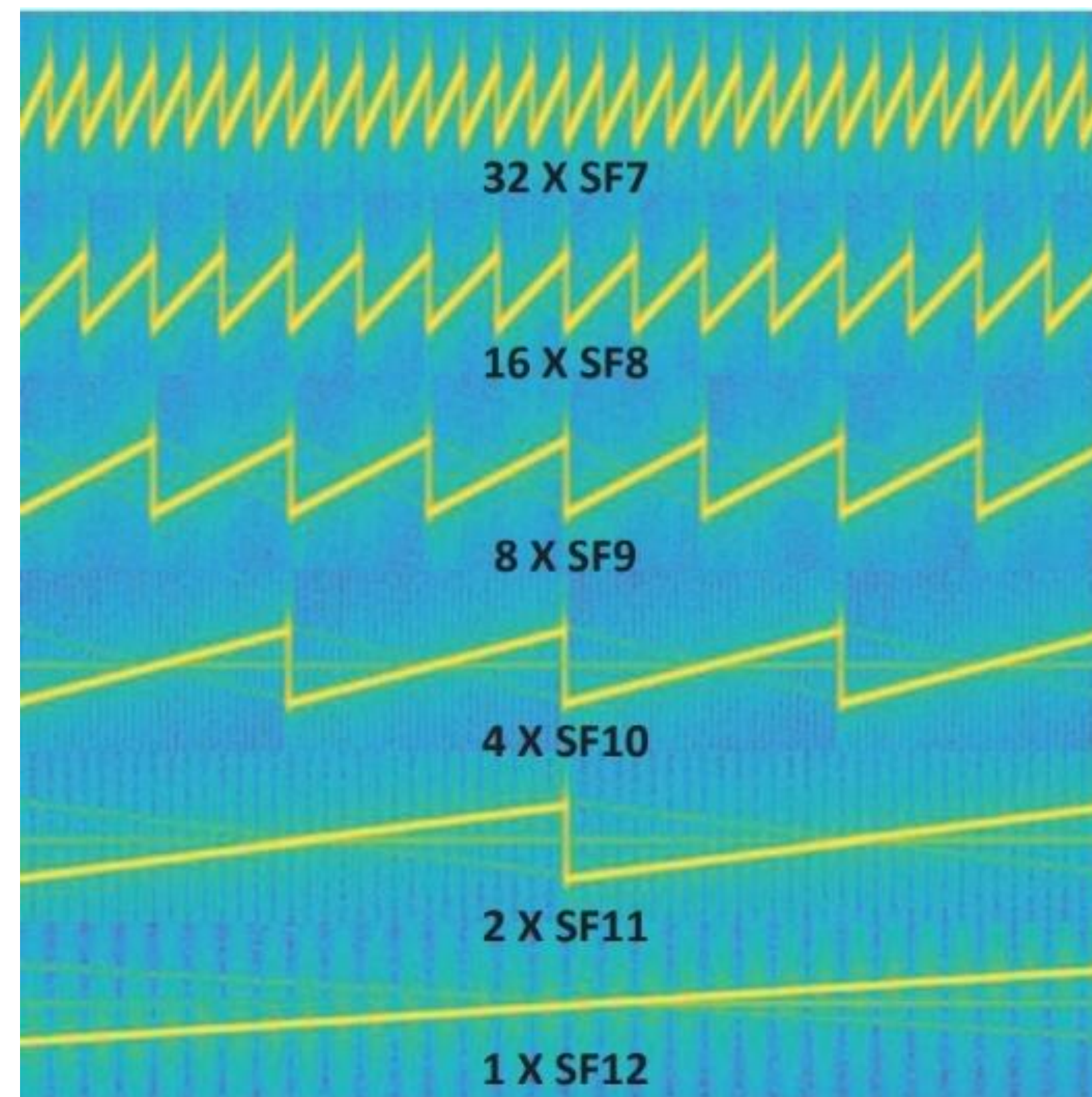
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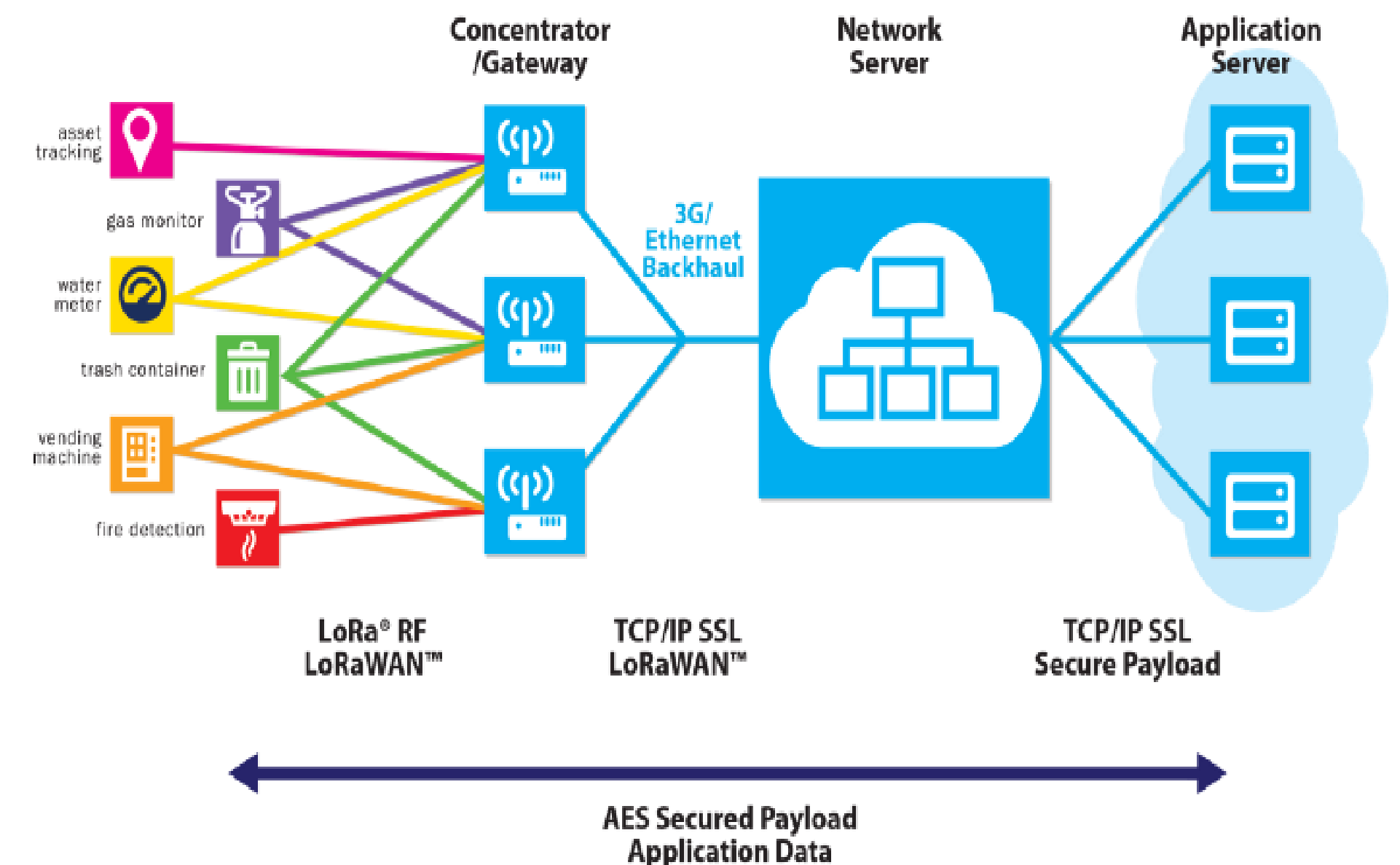
$$N_{payload} = 8 + \max \left(\text{ceil} \left(\frac{8PL - 4SF + 28}{4 \times SF} \right) \times (CR + 4), 0 \right)$$

$$T_{payload} = N_{payload} \times T_{symbol}$$



LoRaWan

Es un protocolo de red de área amplia y baja potencia basado en la tecnología LoRa. LoRaWAN se sitúa en una capa más alta del modelo OSI y se encarga de establecer la estructura de la red, manejar la autenticación y la seguridad, así como coordinar la comunicación entre los dispositivos finales y los servidores de red. LoRaWAN permite la conectividad y gestión de dispositivos IoT a través de redes de área amplia y proporciona un marco estándar para la implementación de soluciones IoT a gran escala.



Outline

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Addressing IPv4

ICMP

IPv6

Conclusion